

Laser scanners with oscillatory elements: Design and optimization of 1D and 2D scanning functions[☆]

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ABSTRACT

Optomechatronic laser scanners with oscillatory mirrors, built as galvanometer scanners (GSs) or as Micro-Electro-Mechanical Systems (MEMS) are approached from the point of view of their scanning functions. The non-linearity of the 1D scanning functions which are the output signals of such scanners (i.e., the current angular position of their oscillatory mirrors) are first discussed for the three most common input signals (triangular, sawtooth, and sinusoidal), for which the effective duty cycle/time efficiency (of the output) was modeled using Optical Coherence Tomography (OCT) imaging, in contrast to the theoretical duty cycle (of the input). Second, optimal linear plus non-linear 1D scanning functions are designed, with specially introduced parabolic portions, in order to maximize the duty cycle of the scanning process and to provide the most distortion-free scanning, for example for OCT. A trade-off is also discussed between this duty cycle and the peaks of the voltage that has to be applied to the motor of the scanner, in order to minimize these peaks and thus to protect the system from an electrical point of view. Finally, using all the results above, both angular and linear 2D scanning functions are designed and analyzed, for their possible variants. For the fast scan axis the triangular 1D functions obtained above (with or without non-linear portions introduced between the constant speed, useful scan) are utilized. For the slow scan axis the two most utilized scanning algorithms are employed: with a step-by-step or with a continuous angular movement. The characteristic parameters of their different variants are analyzed in order to optimize these scanning functions. We demonstrate that the step-by-step angular movement is the best one, in order to avoid producing acceleration (therefore inertia torque and voltage) spikes. Equations that avoid such issues are proposed. These developments are placed in the context of the requirements of high-end applications like OCT, both for biomedical imaging and for Non-Destructive Testing (NDT) in industry, including for hot topics, such as handheld or endoscope scanning probes.

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1. Introduction

Laser scanners are one of the most utilized optomechatronic uni-dimensional (1D) and two-dimensional (2D) laser scanning systems [1,2]. They are employed in a wide range of applications, from commercial (e.g., for barcode scanning or

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printing) to industrial (from optical metrology to laser manufacturing) and to high-end ones, such as Optical Coherence Tomography (OCT) [3,4], Confocal Microscopy (CM), or combinations of such techniques [5].

About forty configurations of laser scanners have been developed since the early 1970s, according to Beiser [2], but only six of them are utilized nowadays:

(i) *Oscillatory*: galvanometer scanners (GSs) or resonant [6–8], the latter usually built as Micro-Electro-Mechanical Systems (MEMS) [9,10]. With regard to all other optical scanners, GSs have good characteristics from all points of view [1,2,7]: very good positioning precisions and fields-of-view (FOVs), good resolutions, reasonable scan speeds and cost per axis, in compact, light-weight constructions. In contrast, all other scanners have only one or two of these characteristics better than GSs, while others are worse. This makes other scanners except GSs mostly suitable for niche applications, as presented in the following.

(ii) *Rotational polygon mirrors (PMs)* [11–15]: they can have high rotational speeds (usually up to 54 krpm, but even towards 70 krpm, lately) [12], therefore, taking also into account their number of facets (of up to 128 [13]), they have much higher scan speeds than GSs. However, they are larger, more expensive, and are affected by mechanical issues due to these high rotational speeds, including vibrations, noise, and mechanical wear [1].

(iii) *Refractive – mostly with Risley prisms*: their scan speeds are also higher than GSs [16], but they have complicated, non-linear, and more difficult to follow scan patterns [17] than the raster scanning produced with dual axis GSs [18] which is the main aim of our study.

(iv) *Holographic*: they solve some of the issues of PMs, but they can be used only with the wavelength their holograms were produced at, therefore their use is limited [2].

(v) and (vi) *Acousto- and electro-optical scanners* have, with regard to all systems above, the advantage of a faster positioning of the laser beam, because of the lack of mechanical inertia [1,2]. However, their resolution is quite low [1], therefore their range of applications does not include high-resolutions techniques, like biomedical imaging, for example.

Taking into account all these aspects, as well as their mature technology (regarding motors, bearings, heat sinkers, mirrors, sensors, control structures, and testing) [7], technology that is still advancing [8], GSs imposed themselves in most applications, especially for lateral scanning, including in rapidly evolving techniques like laser manufacturing and Non-Destructive Testing (NDT) in industry [19–22], as well as OCT, CM, two photon or multi-photon microscopy, for example, in biomedical applications [3–5,9,10,23–25].

However, GSs still have issues, especially regarding their scanning functions and algorithms [26–31], but also their control structures [32–34]. Also, as the tendency nowadays is towards miniaturization and as most MEMS scanners are built with oscillatory micro-mirrors, the progress achieved regarding these aspects is valid for such MEMS, as well [9,10]. What is more, the domains of GSs and MEMS are increasingly overlapping, as the minimum aperture of the former and the maximum one of the latter are both in the range of 3–4 mm. On the other hand, GSs are also growing in size, as needed in laser manufacturing, for example, while MEMS are also downsizing towards sub-millimeter aperture, as required in hot topics like the construction of endoscope probes, therefore the range of constructive parameters of oscillatory scanners increase.

This provides an active avenue of research for the optimization of their scanning functions, algorithms, and programming, as we have approached, both theoretically and experimentally – for higher duty cycles/time efficiencies and low distortions – in OCT imaging [26,28,30]. Another research avenue, on their control structures, focuses on reaching higher response speeds (for real time, in vivo biomedical imaging or in situ NDT in industry) and a better stability to perturbations – for an appropriate use in both clinical and industrial environments [32–34]. The latter aspect has also been leading to a strong development of handheld and endoscope scanning probes [35–40]. We point out such advancements in the present work, related to their requirements for GSs and MEMS, but the main aim of this study is to bring under the same umbrella our already-completed modeling on 1D scanning using common input signals, i.e. triangular, sawtooth, and sinusoidal (utilizing our experimental works on scanning in OCT [26,28,30,31]), to optimize 1D scanning functions with custom-made input signal (including from the point of view of the voltage peaks applied to the scanner), initially developed in [30], as well as to apply all of these in the analytical development of 2D scanning functions, with a focus on high-end applications. A multi-disciplinary, optomechatronic approach is utilized, combining the optical requirements of laser scanners with their analysis and optimization from a mechanical point of view (as related to their oscillatory movement), but also to automation and electrical aspects, the latter regarding the voltage peaks of the driving motors of scanners, and their control structures. A brief presentation of these contributions has been made recently in [41].

The remaining of this paper includes in Section 2 a discussion of the modeling of the effective scanning functions/output signals of laser scanners with oscillatory elements, for their most common input signals. Pre-shaped input signals are approached in Section 3 in order to obtain the highest possible duty cycle/time efficiency of the scanning process for such systems. Their optimization is discussed, for OCT applications and devices, from the point of view of their necessary performances and of their fulfillment. Section 4 concludes the discussion on 1D GSs with their command functions, in order to reduce the electrical inputs. In Section 5, optimal 2D scanning functions are obtained, based on the above discussions – for both 2D scanners with dual axis GSs or with MEMS with oscillatory micro-mirrors. Section 6 concludes the study and provides directions of future work.

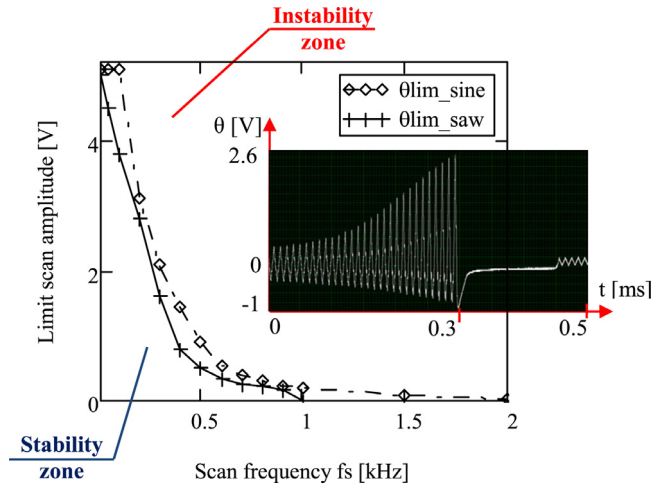


Fig. 2. Limit angular scan amplitude of a GS – with regard to the scan/oscillatory frequency of the galvomirror, for two scanning functions: triangular and sinusoidal.

to the effective one, t'_a) and in rotational angle (from the scan amplitude of the input, θ_m , to the one of the output, θ_a) – Fig. 1.

There are thus *theoretical parameters of the scanning process*, that characterize the input signals, including t_a and θ_m , as well as the linear amplitude H , but also the *theoretical/ideal duty cycle* η_t (of the input signal), which we defined in [28] as

$$\eta_t = \frac{t_a}{T} = 1 - \frac{\Delta t}{T}, \quad (1)$$

where Δt is the (theoretical) return time for the sawtooth input function – Fig. 1(c1). For the triangular input function this time interval $\Delta t = T/2$, Fig. 1(b1). In order to make the comparison, a one-directional scan was considered in both cases; for the most useful (and utilized) two-directional scan in the case of triangular input functions, $\Delta t = 0$, Fig. 1(b1).

These theoretical parameters differ from the *effective parameters of the scanning process*, that are those of the output signals, including t'_a and θ_a , as well as the effective (useful) linear amplitude x_a (scanned with $v = \text{cst.}$), but also the *effective duty cycle* η (of the output signal), which is

$$\eta = \frac{t'_a}{T} = 1 - \frac{\Delta t'}{T}. \quad (2)$$

One can see from Fig. 1(b2) versus Fig. 1(b1), as well as from Fig. 1(c2) versus Fig. 1(c1) the differences between the output/effective and the input/theoretical signals – for triangular and sawtooth scanning, respectively. These can be concluded from the detailed experimental study carried out in [26,28] on GSs in high-end applications i.e., OCT. In brief, other aspects demonstrated experimentally in these studies – and which are useful in the present one – are:

(i) *Sinusoidal scanning* is the most advantageous from a mechanical point of view, as it corresponds to the smoothest oscillatory regime. This can be remarked from the lack of vibration and noise (that are present in sawtooth scanning, for example), but also from the study of the limits of the (angular) scan amplitude θ_{lim} with regard to the scan frequency f_s [26]. One can thus remark that the maximum possible f_s for sinusoidal scanning is double than the one for triangular scanning, i.e., 2 kHz versus 1 kHz, for a 5 mm aperture GS. Such limits of the maximum possible f_s for these two scanning functions are provided by GS manufacturers, but not the type of graph given in Fig. 2. However, knowing this boundary between the stability oscillatory regime and the instability one (for which chaotic vibrations appear) is necessary for GS users. From an optical point of view, the sinusoidal scan has a major drawback, as it implies a non-constant scanning speed for the entire oscillatory period. In imaging for example, this means that the pixels in OCT or CM are not equally spaced, therefore the image is distorted [26,38]. An approximation can be made and the useful scan could be reduced to the portion around the O.A., but this means sacrificing both the effective duty cycle and the FOV.

(ii) *Triangular scanning* is the best one from the point of view of obtaining the highest possible duty cycles and the smallest distortions, situated only at the margins of the scanned field, as demonstrated experimentally in [26] using OCT. The mechanical regime of triangular scan is however not as smooth as the sinusoidal one, as shown in Fig. 2.

(iii) *Sawtooth functions* provide a one-directional scan, that is an advantage in certain applications, but they have mechanical issues, including vibrations, asymmetry of the working regime, heating, and wear. They thus affect both the functioning and the life expectancy of the scanner. Also, the return portion (that has only theoretically the duration $\Delta t = (1 - \eta_t)T$, and effectively the larger duration $\Delta t'$ – Fig. 1) has to be discarded.

Using this study, another group has applied sawtooth scanning [27], but discarding the distorted portions evaluated based on [26], of a duration $\Delta t'$ – Fig. 1(c2), in order to obtain collated images of the retina using Doppler OCT. Such images are

necessary because, in OCT, for example, in order to have a high lateral resolution, small areas of the investigated samples, of even $0.5 \times 0.5 \text{ mm}^2$ are imaged. As samples are usually larger (e.g., for retina, cornea, teeth, skin, endoscopy, or in NDT), individual OCT images must be collated to obtain the entire image. As for high scan frequencies and amplitudes distorted bands appear at the margins of individual images, as discussed above, the large image is affected by a grid of blurred 'lines' that may limit the investigations or produce false responses (the latter for biomedical samples). Taking into account this necessity to obtain collated images, larger than for the FOV of individual OCT images, we further developed in [28] a theory of the effective duty cycle for sawtooth scanning (which includes triangular scanning as a particular case when η_t equals 50%).

In [28] we first studied the effective duty cycle η with regard to the theoretical one η_t , at different levels of the scan frequency and amplitude. A phenomenon was thus determined, for the first time to our knowledge: the oscillatory scanner actually saturates, i.e., if η_t is chosen too high (e.g., towards 90%), η is expected to follow it, but it does not; it actually decreases, while its maximum value is reached for a η_t of around 75%. The phenomenon does not happen at low f_s (roughly under 100 Hz); it strongly manifests as f_s is increased (at 200–300 Hz), especially at high scan amplitudes/FOVs, while at too high f_s (>500 Hz), η decreases severely for any scan amplitude. The relevance of this phenomenon and remarks is given by the fact that the second range of f_s above is the most utilized one in high-end applications like OCT, and it is the one considered in the following sections.

The equations of the effective duty cycle η are, using Eq. (2) [28]:

$$\eta = \begin{cases} \frac{1}{2} - \frac{2\tau}{T}, & \text{for triangular scan} \\ \eta_t - \frac{\tau_1 + \tau'_1}{T}, & \text{for sawtooth scan} \end{cases} \quad (3)$$

where the effective return time of the galvomirror is $\Delta t' = 2\tau$ for the triangular scan, Fig. 1(b2), as well as $\Delta t' = \tau_1 + \tau'_1 + \tau_2 + \tau'_2$ for the sawtooth scan, Fig. 1(c2). As pointed out previously, these time intervals are increased with regard to those that characterize the input signal: for the triangular scan $\Delta t = T/2$ for the one-directional scan in Eq. (3), Fig. 1(b1), while for a two-directional scan, as $\Delta t = 0$, η is double than in Eq. (3) ₁; for the sawtooth scan, Δt is the theoretical flyback in Fig. 1(c1).

These equations, obtained by modeling the GS output, were also validated by measuring directly the distortions in the OCT images [28], using the state-of-the-art 2 μm axial and lateral resolution Gabor Domain Optical Coherence Microscopy (GD-OCM) [29] (in contrast to the more usual 10 μm resolution OCT [4]). Modeling the distortions at the margins of OCT images (but the same technique can be applied for CM, for example) led to the development of algorithms to discard these distorted portions by the partial overlapping of individual images, and thus to obtain larger FOV, i.e., collated images without artifacts [28].

As a remark, we eliminated with this discussion both the sinusoidal scanning regime, which is characterized by significantly distorted images [26], and the sawtooth scanning with a combination of a too high scan frequency and amplitude, which we shown to shifts towards a triangular-like shape, but a deformed one, with none of the advantages of the triangular input and with all the mechanical issues of the sawtooth one [28]. From the point of view of the scanning functions, the triangular ones are therefore considered in the remaining of this paper.

From the point of view of the scanner itself, 1D GSs are more rarely utilized in this form; more often applications utilize 2D GSs or MEMS, with two perpendicular oscillatory axis. The reason for this is that a 1D GS provide only a scan line (and for OCT, a B-scan, i.e., a cross-section of the sample), while a 2D GS or MEMS scanner provide the scanning of a plane (and for OCT, a volumetric reconstruction of the scanned portion of the sample). For OCT, in order to move the technology from the laboratory to clinic (for biomedical imaging) and in situ for NDT [22], mobile OCT units are made, equipped with handheld scanning probes [35–40].

A hot topic nowadays thus refers to developing such probes, with different laser scanners, as appropriate for certain applications. We have designed in this respect a family of OCT probes equipped with 1D GSs [37] that are capable to produce B-scans/transversal images into the sample, as it is sufficient in dental medicine, where the dentist provides anyway the second scan axis by sweeping the area of interest on the hard or soft tissue with the handheld probe. Investigations of the oral chords, the larynx, and the tympana were also made with such probes, already installed in the Ear–Nose–Throat (ENT) Department of the Northwick Park Hospital (London, UK). On the other hand, the most utilized – and necessary – are scanning probes equipped with 2D scanners, which allow for volumetric reconstructions. Mostly 2D GSs are used for such probes, because of their superior FOV [35,40], but MEMS-based probes are also in full development [38,39]. This is why a study on scanning functions cannot be limited to 1D functions; the development of 2D scanning functions becomes essential – as approached in Section 5.

3. Pre-shaped input signals to optimize the duty cycle of 1D scanners

In contrast to the common input signals approached in the scanning analysis in the previous section, in order to increase the duty cycle of a GS (but this discussion is also valid for MEMS scanners), the literature points out the possibility to use input signals with linear portions for the active scans (i.e., for the constant speed portions), but with pre-programmed non-linear portions placed in-between these useful scans [1,6,7]. Taking into account the discussion in the previous section, this

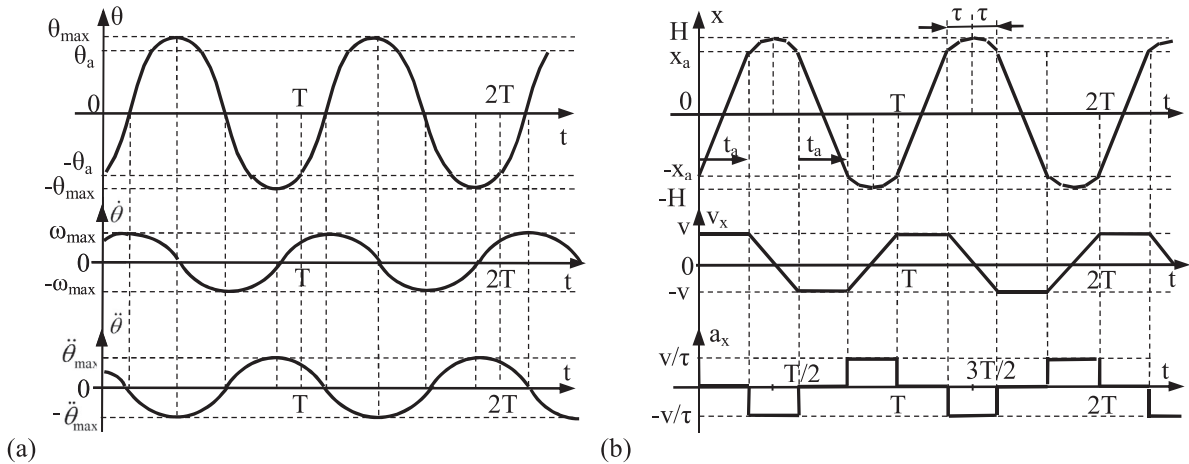


Fig. 3. (a) Angular scanning function of a 1D GS, $\theta(t)$, with its derivatives (angular speed $\dot{\theta}$ and acceleration $\ddot{\theta}$) and (b) linear plus parabolic (p) function $x(t)$ produced by this angular function, with its derivatives (speed v_x and acceleration a_x).

is the appropriate method [26,28]: it gives the system a necessary 'break' to avoid it from becoming saturated, otherwise the scanner produces (because of the mechanical inertia) its own non-linear portions for the scanning functions.

Two main input signals have been pointed out in the literature as potential candidates for this purpose: linear plus parabolic (p) or linear plus sinusoidal (s) [1,6,7]. Furthermore, the statement in the literature has been that the optimal signal (i.e., the one that produces the highest effective duty cycle η) is linear plus sinusoidal (s) [7]. However, as this was contradicting our observations, we have begun to approach in [30] this problem for GSs in open loop. However, one has to mention that GSs can be driven in open loop (and so can MEMS scanners with oscillatory micro-mirrors), but such devices are only used in special (niche) applications. Therefore, using also our study in [34], on control structures of GSs, the aim of this section is to extend the discussion made initially for open loop GSs for the most utilized case of closed-loop devices, in order to obtain optimal 1D scanning functions (that are also necessary to further develop 2D scanning functions – Section 5).

The movement equation of the oscillatory element of the GS (which includes its galvomirror – Fig. 1) in open loop is the classical one:

$$J\ddot{\theta} + c\dot{\theta} + k\theta = T_a(t), \quad (4)$$

where J is the inertia moment of the mobile element (with regard to its rotational axis), k is the elastic coefficient of its spring, c is the damping coefficient, and $T_a(t)$ is the active electro-magnetic torque (Fig. 1). Eq. (4) can also be written as:

$$\ddot{\theta} + 2\xi\omega_0\dot{\theta} + \omega_0^2\theta = T_a(t)/J, \quad (5)$$

where $\omega_0 = \sqrt{k/J}$ and $\xi = c/2\sqrt{Jk}$ are the undamped angular frequency and the damping ratio of the system, respectively.

For GSs or MEMS utilized in high-end applications like biomedical imaging (e.g., OCT or CM) [5], closed-loop scanners have to be used, in order to control the position of the oscillatory mirror at all times. We obtained the transfer function of scanners of such GSs equipped with the most common proportional-derivative (PD) controllers [34], but also with more advanced control structures (including predictive), in order to obtain a higher response speed or a better stability to perturbations. Thus, for the GS parameters that we have experimentally deduced and for their useful range of oscillatory frequencies (Fig. 2), we demonstrated, for the first time to our knowledge, that the transfer functions of such devices can be approximated with a second order one [34]. The movement of the oscillatory element of closed-loop GSs therefore obeys the classical second order Eq. (5). A similar approach as for open loop GSs can thus be made for closed-loop GSs and MEMS, as well.

Fig. 3 presents the shape of the angular and linear scanning functions, $\theta(t)$ and $x(t)$, respectively, for the linear plus parabolic (p) function, as well as their derivatives. One has to remark that the linear plus sinusoidal (s) function looks similar to the (p) one; only their derivatives are different. Regardless of the equation of the linear scanning function, the angular one is non-linear on the entire time period T (therefore it has a similar shape – Fig. 3(a)), as the relationship between them is, from Fig. 1,

$$\theta(t) = +\frac{1}{2} \arctan \frac{x(t)}{L}, \quad (6)$$

where, if an objective lens is used, the distance L between the pivot of the galvomirror and the principal object plane of the lens is equal to its object focal distance f .

In the detailed discussion in [30] several possible scanning functions were considered, including linear plus higher order polynomials and linear plus sinusoidal-type functions with a more complicated equation. By doing this, we were able to

eliminate them from the discussion by demonstrating that all these functions produce much lower duty cycles than the (p) and (s) functions above. The (p) and (s) functions are therefore indeed the potential candidates for the GSs' highest duty cycle, as stated in the literature.

Their equations are obtained in the Appendices, by taking into account the conditions from Fig. 3(b) for $x(t)$, which we pointed out that has the same shape for (p) and (s). From Appendix A, using Eqs. (A.2), (A.5), (A.8), and (A.10), the linear plus parabolic scanning function (p) is

$$x(t) = \begin{cases} vt - x_a, & \text{for (i) } t \in [0, t_a) \\ \frac{v}{2\tau} [-t^2 + 2(t_a + \tau)t - t_a(t_a + \tau)], & \text{for (ii) } t \in [t_a, T/2) \\ v\left(-t + \frac{3t_a + 4\tau}{2}\right), & \text{for (iii) } t \in [T/2, T - 2\tau) \\ \frac{v}{2\tau} [t^2 - 2(T - \tau)t + T(T - 2\tau)] + x_a, & \text{for (iv) } t \in [T - 2\tau, T) \end{cases}, \quad (7)$$

From Appendix B, using Eqs. (B.8) and (B.11), the linear plus sinusoidal scanning function (s) is

$$x(t) = \begin{cases} vt - x_a, & \text{for (i) } t \in (0, t_a) \\ H \cos \omega(t + T/2 + \tau), & \text{for (ii) } t \in [t_a, T/2] \\ v\left(-t + \frac{3t_a + 4\tau}{2}\right), & \text{for (iii) } t \in [T/2, T - 2\tau] \\ -H \cos \omega(t + \tau), & \text{for (iv) } t \in [T - 2\tau, T] \end{cases}, \quad (8)$$

where ω is given by Eq. (B.4) and τ by Eq. (B.5). The duty cycle of the GSs is therefore obtained, from Eq. (2), using Eq. (A.7) for the (p) function, and Eq. (B.5) for the (s) function:

$$\eta = \frac{t_a}{t_a + 2\tau} = \begin{cases} \frac{r}{2-r}, & \text{for (p)} \\ \frac{1}{1 + \frac{\sqrt{1-r^2}}{r} \arctan \frac{\sqrt{1-r^2}}{r}}, & \text{for (s)} \end{cases}. \quad (9)$$

From these two equations, the duty cycle η_s – for (s) is slightly higher than η_p – for (p), when studied with regard to the ratio $r = x_a/H$ (of the effective/useful and maximum/set scanning amplitudes – Fig. 1), ratio which is somehow similar to the duty cycle, but in space, not in time. This may be a reason for the statement made in the literature [7], that the (s) function is better than the (p) function. However, in order to determine the best scanning function, one also has to take into account the maximum inertia torque for (p) and (s) [30]. From Eq. (4) and from Fig. 1

$$T_{\max} = J\ddot{\theta}_{\max} = J\ddot{\theta}(t_a + \tau) \quad (10)$$

where, using Eq. (6), the angular acceleration is

$$\ddot{\theta}(t) = \frac{\ddot{x}}{2L[1 + (x/L)^2]} - \frac{x\ddot{x}^2}{L^3[1 + (x/L)^2]^2}. \quad (11)$$

As from Fig. 3, $\dot{x}(t_a + \tau) = 0$, from Eqs. (10) and (11),

$$T_{\max} = \frac{J|\ddot{x}_{\max}|}{2L[1 + (x/L)^2]}. \quad (12)$$

Therefore, by using Eq. (A.7) for the (p) function and Eq. (B.4) for the (s) function,

$$T_{\max} = \begin{cases} \frac{Cv^2}{1-r}, & \text{for (p)} \\ \frac{2Cv^2}{1-r^2}, & \text{for (s)} \end{cases} \quad (13)$$

where the following notation was used:

$$C = \frac{J}{4HL[1 + (H/L)^2]} \quad (14)$$

A normalized, a dimensional torque can thus be obtained:

$$T_i = T_{\max}/Cv^2 \quad (15)$$

which, taking into account Eq. (14), only depends on the ratio r – for each scanning function. One can easily verify that $T_{is}(r) > T_{ip}(r)$, therefore, for a certain GS (characterized by the parameters J , L , H , and v , therefore by a certain value of the

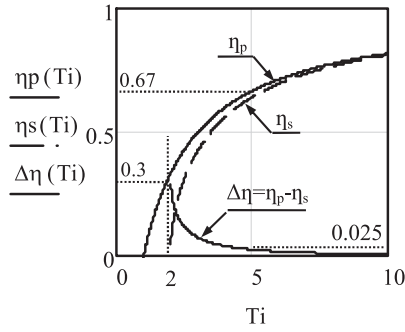


Fig. 4. Duty cycles η_p and η_s of the linear plus parabolic (p) and linear plus sinusoidal (s) scanning functions, respectively, as well as their difference $\Delta\eta = \eta_p - \eta_s$.

coefficient Cv^2), a higher ratio r can be reached with the (p) function. Algebraically, by combining Eqs. (9) and (13), taking also into account in the latter Eq. (15), and by eliminating the ratio r for each scanning function, the final expressions of the duty cycle can be obtained, similar to [30]:

$$\eta = \begin{cases} (T_i - 1)/(T_i + 1), & \text{for } (p) \\ \left(1 + \sqrt{\frac{2}{T_i - 2}} \arctan \sqrt{\frac{2}{T_i - 2}}\right)^{-1}, & \text{for } (s) \end{cases} \quad (16)$$

The final comparison of the two scanning regimes is made in Fig. 4. The conclusion is that $\eta_p > \eta_s$ for any GS, therefore the linear plus parabolic function (p) produces the highest possible duty cycle. The difference $\Delta\eta = \eta_p - \eta_s$ is also given in Fig. 4 to see if it has useful values. One may thus remark that it has a maximum value of 30% for $T_i = 2$ (for which η_p equals 30%, while η_s equals 0). However, for higher values of T_i , $\Delta\eta$ grows smaller, as the two graphs η_p and η_s are getting closer. This difference is however still significant for fast scanning applications, being equal to 2.5% at η_p equals 67%, therefore around the most used levels of the duty cycle.

This type of optimization of GS or MEMS scanners can be utilized in any application that requires a distortion minimization and an increase of the duty cycle of the scanning process, from laser manufacturing to biomedical imaging. A relevant example for the latter is OCT, for which the above results were exploited in achieving the first real-time, distortion-free OCT imaging (without post-processing) with a prototype MEMS-based handheld probe and with the high 2 μm axial and lateral resolution GD-OCM [29]. In state-of-the-art MEMS-based OCT probes, resonant devices, therefore sinusoidal scanning were employed [38]; this produced large distortions in OCT images, therefore extensive post-processing had to be used, and real-time (thus, in vivo) imaging was not possible. The solution found in [39] was to use triangular scanning, based on its demonstrated capability to produce the most distortion-free OCT image at the highest duty cycle [26]. In order to do that, the MEMS scanner had to be used in open loop, therefore the triangular input was affected by small distortions throughout the entire period T , distortions that had to be filtered. The result was precisely a linear plus non-linear scanning function of the type deduced in this section. However, as the non-linear portions of the signal employed in [39] have not been (p) or (s) (but rather a function in-between them), further improvement may be necessary in this respect, in order to assure not only distortion-free, but also maximum duty cycle scanning.

4. Command functions of oscillatory scanners

The command function of the GS or MEMS scanner is obtained by replacing in Eq. (4) the expressions of the derivatives $\dot{\theta}(t)$ and $\ddot{\theta}(t)$ obtained from Eq. (6). The active torque therefore has the expression:

$$T_a(t) = \frac{-Jx\ddot{x}^2}{L^3[1 + (x/L)^2]^2} + \frac{J\ddot{x} + c\dot{x}}{2L[1 + (x/L)^2]} + \frac{k}{2} \arctan \frac{x}{L}. \quad (17)$$

On the other hand, $T_a(t)$ is proportional with the input voltage and with the magnetic flux that passes through the mobile element of the electrical motor – Fig. 1(a). Therefore Eq. (17) actually provides the command function of the system, which is the necessary voltage input required to obtain the scanning function $x(t)$ [31].

To analyze the peaks (i.e., the maximum reached values) of this voltage/torque function, a convenient change of the origin of the time axis in Fig. 3 has to be made, to bring these peaks at $t = 0$. This way, one can make an easy comparison of the values of these peaks; otherwise they would be placed at different time moments, depending on the oscillatory time period (i.e., on the scan frequency f_s) of the $x(t)$ scanning function. Using a translation of the time origin with $t_a + \tau$, in Fig. 3(b), the equation of this first parabolic portion can be easily deduced (similar to Appendix A) as:

$$x(t) = \frac{-vt^2}{2\tau} + H, \text{ for } t \in [-\tau, \tau]. \quad (18)$$

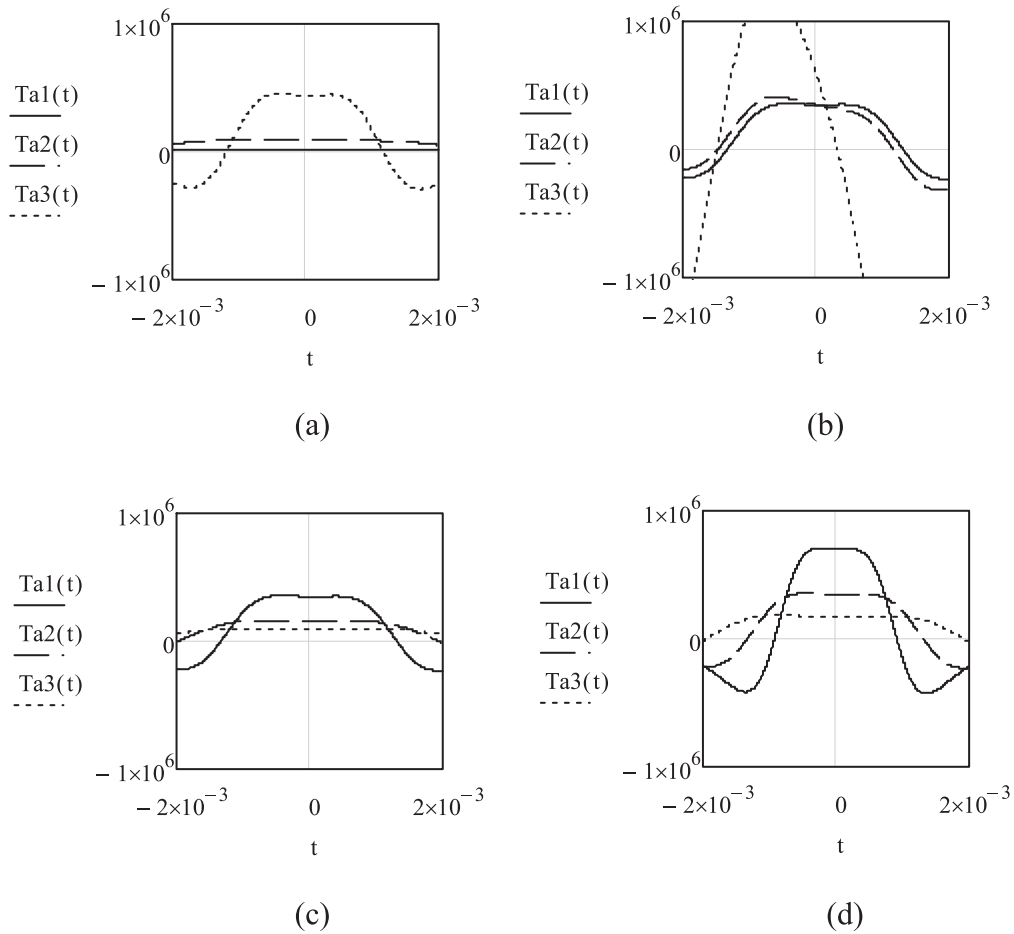


Fig. 5. The normalized active torque of a GS for $L = 200$ mm, and for: (a) three scan frequencies $f_{1s} = 50$ Hz, $f_{2s} = 100$ Hz, and $f_{3s} = 200$ Hz, with $H = 20$ mm and $x_a = 19$ mm (therefore $r = 0.95$); (b) three natural frequencies of the mobile element: $\omega_{01} = 10$, $\omega_{02} = 150$, and $\omega_{03} = 2250$ rad/s; (c) for the same $H = 20$ mm, but for three ratios r (therefore for three duty cycles): $r_1 = 0.95$ ($\eta_3 = 90.48\%$), $r_2 = 0.89$ ($\eta_2 = 80.18\%$), and $r_3 = 0.82$ ($\eta_3 = 69.49\%$); (d) for three scan amplitudes of the input signals: $H_1 = 10$ mm, $H_2 = 20$ mm, and $H_3 = 40$ mm – for the same ratio $r = 0.95$ (therefore $\eta = 90.48\%$), therefore the scan (useful) amplitude (of the output signal) is $x_{1a} = 9.5$ mm; $x_{2a} = 19$ mm, and $x_{3a} = 38$ mm, respectively.

This equation will be considered in the following analysis, in Eq. (17). As a remark, this simpler form cannot however be used for the scanning function, as one does not want the scanning to begin with a non-linear, but with an active, linear portion, as in Eq. (7).

The parameters of the scanning process (unless otherwise specified) are: $L = 200$ mm, $H = 20$ mm, and $x_a = 19$ mm, therefore the ratio $r = x_a/H$ equals 0.95. The duty cycle, the time interval 2τ of the non-linear portions, and the scanning speed v are determined by using the designing Eq. (A.6) obtained in Appendix A for the (p) scanning function. Thus, for this value of r , $\eta = 90.48\%$ (which is around the highest possible value for most scanners). For the scan frequency $f_{1s} = 50$ Hz, $\tau_1 = (1-\eta)/4f_s = 119$ μ s and $v_1 = 193.97$ m/s; for $f_{2s} = 100$ Hz, $\tau_2 = 247.75$ and $v_2 = 8,387.94$ m/s; for $f_{2s} = 100$ Hz, $\tau_3 = 381.33$ μ s and $v_3 = 16,775.88$ m/s. The values $\omega_0 = 10$ rad/s and $\xi = 1$ (therefore critical damping) are also considered.

The peaks of $T(t)$, proportional to the peaks of the input voltage for the GS, are the main issue that has to be discussed, in order to protect the electrical circuit. They have to be decreased to a value that GSs can whist and from the point of view of the input voltage (therefore of the active torque), in order to obtain feasible devices from both an electrical and a mechanical point of view. Making an initial analysis, the peaks of the non-linear portions of the scanning function are the most significant, as they are with two orders of magnitude higher than for the linear portions. Also, from Fig. 5(a) these peaks increase roughly by a factor of four when the scan frequencies are increased in the progression we considered: from 50 to 100, and to 200 Hz. One may also conclude from such a similar analysis (not included here) that this increase is also significant for lower values of f_s : thus the peaks grow by a factor of two for each increase of the f_s from 10 to 20, and to 50 Hz.

Several methods can be considered in order to avoid voltage peaks that would actually affect the GS – even if theoretically they seem to provide the optimal scanning function (p) obtained in the previous section:

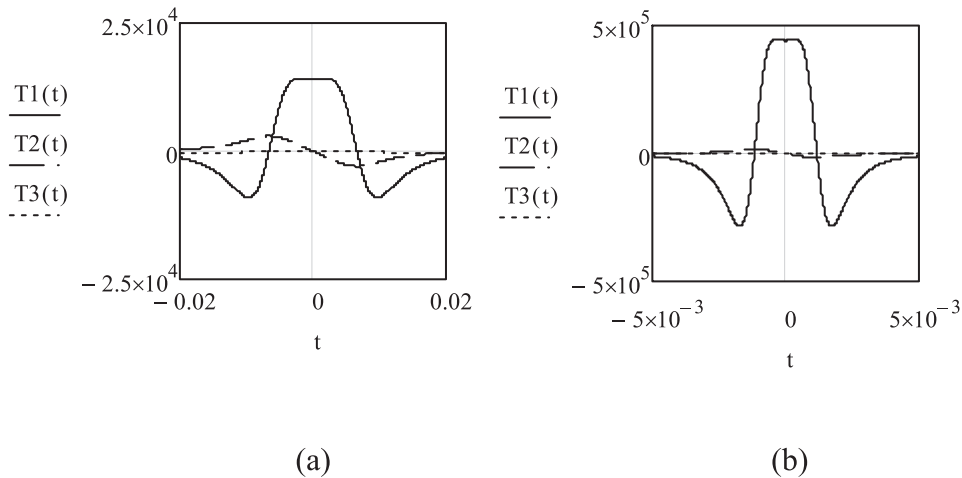


Fig. 6. Components of the active torque for $L = 200$ mm, $H = 20$ mm, $x_a = 19$ mm: T_1 , inertia torque; T_2 , damping torque; T_3 , elastic torque (adimensional); comparison for two scan frequencies f_s : (a) 20 Hz and (b) 200 Hz, therefore (a) $v = 3356$ m/s and (b) $v = 18879.77$ m/s – for each time periods specific to these frequencies.

- (1) By adjusting the parameters ω_0 and ξ – Eq. (5): in Fig. 5(a), $T(t)$ is studied with regard to ω_0 . As ω_0 is a function of J , it depends on the size and mass of the mobile element, which includes the shaft and the galvomirror. However, these are imposed, for a certain application, by the scanner aperture, as well as by the power of the laser utilized. One may conclude from Fig. 5(b), where three values, in geometrical progression, were considered for ω_0 , that only high values of ω_0 produce an increase of the torque/voltage peaks. This is the case of small inertia, i.e. for small aperture GSs or for the micro-mirrors of MEMS scanners. As a remark, the other parameter from Eq. (5), ξ has an impact that can be neglected on the active torque, because of its closed values.
- (2) By giving up some of the duty cycle (i.e., by not increasing it to the maximum): as seen from Fig. 5(b), by doing this, the voltage peaks can be reduced roughly to a half with every decrease of the ratio $r = x_a/H$ that was considered in Fig. 5(b), therefore with an approximate 10% reduction of η . From Fig. 5(b), by considering an f_s of 200 Hz, the torque/voltage is thus decreased to more acceptable levels: from η_1 to η_2 the peaks decrease to less than half the value, while from η_2 to η_3 and further on, this decrease is less significant. One may conclude that η_2 is for this f_s the right choice, and such an analysis can be performed for every different set of scanner parameters. The actual η at which a scanner should be used – even when using the theoretically optimum scanning function (p) – can thus be determined.
- (3) By preserving the duty cycle, but increasing the scan amplitude H that corresponds to the angular scan amplitude θ_m of the GS's input signal – Fig. 1. From Fig. 5(c), one may see that the effect is similar to the previous method. The peaks of $T(t)$ thus decrease roughly by a factor of two, for each step that was considered for the decrease in H . This can be explained by the fact that, for the same η , the galvomirror has more space to stop-and-turn. The problem is that H is related to the FOV of the device, which usually should be as large as possible (and it is anyway a requirement of the specific application). On the other hand, the maximum/limit angular scan amplitude θ_{lim} is a function of the scan frequency (Fig. 2) – in order to assure stable oscillations. A solution to address these issues is to increase (if possible, and limited by the dimension of the setup), the distance L (therefore the focal distance of the objective lens – Fig. 1). It is thus possible to increase H for a given angular scan amplitude θ_m . Therefore the necessary value of x_a can still be used, r can be reduced, η can be more or less preserved, and the peaks of the torque/voltage can be significantly decreased.

The final step in this section of the study is to determine the value of the duty cycle η that can actually be achieved for an imposed scanning speed v , in order to reach only a maximum (allowed) value of the active torque $T_a(t)$. To complete this, in Fig. 6 the three components of the active torque, Eq. (17), are compared for the considered time interval, for the parabolic portion, Eq. (18) – for two values of the f_s . One can see that the value of the inertia component of the torque, T_{a_max} is much higher than the other two components, i.e., the damping and the elastic torque – especially for the higher f_s – therefore for this time interval T_a can be approximated with T_{a_max} .

From Eq. (13), using the (p) scanning function, Eq. (7), the maximum duty cycle of the oscillatory scanner that can be reached for a certain T_{a_max} and v is:

$$\eta_{max} = \frac{T_{a_max}/Cv^2 - 1}{T_{a_max}/Cv^2 + 1}, \quad (19)$$

where C is the coefficient given by Eq. (14).

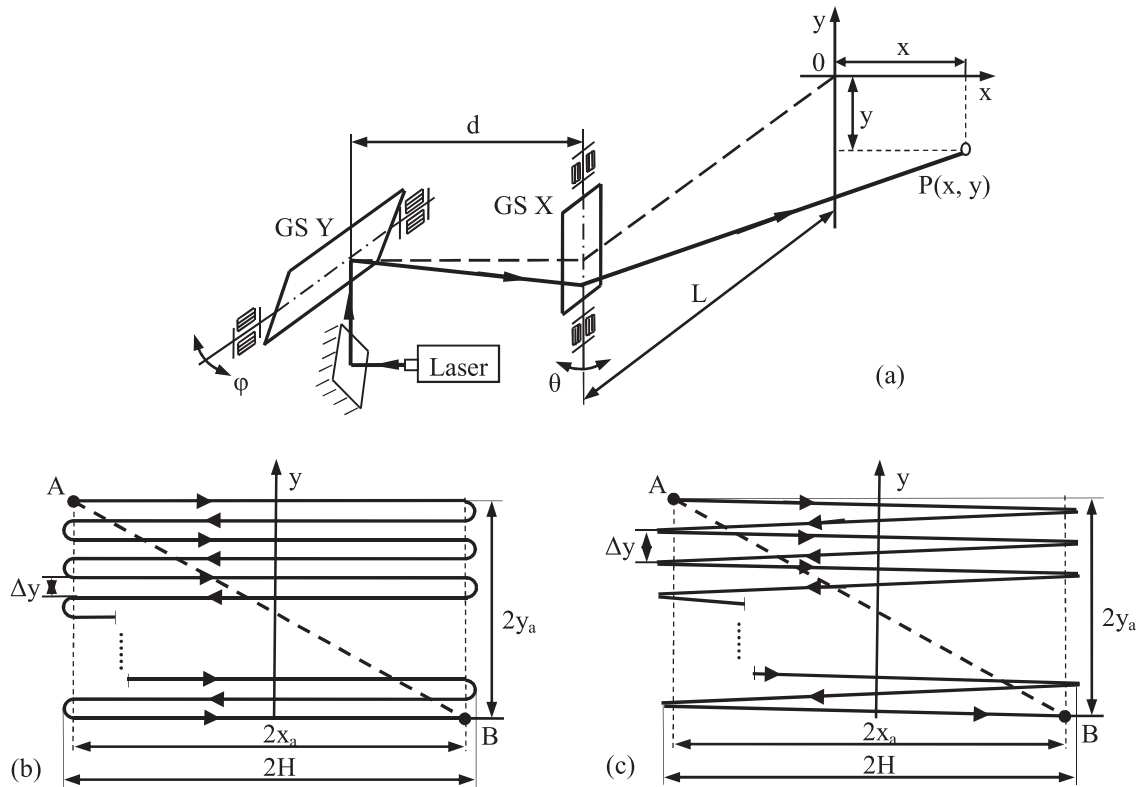


Fig. 7. (a) 2D laser scanning system with two GSs with perpendicular axis (GS X for the fast axis scan and GS Y for the slow axis scan); (b) raster scanning with step-by-step angular movement on the slow axis Oy; (c) raster scanning with a constant linear speed on the slow scan axis Oy.

5. Design and optimization of the scanning functions of 2D GSs and oscillatory MEMS

5.1. 2D scanners and algorithms

A 2D dual axis laser scanning systems with two GSs with perpendicular axis has the following geometrical parameters – Fig. 7(a): L , the distance from the axis of fast axis scanner (the GS that provides the x -scan, GS X) to the Oxy screen/scanned plane (or, in some cases another optical element, such as the objective lens in Fig. 1, or mirrors [42]); d , the distance from the axis of the slow axis scanner (the GS that provides the y -axis scan, GS Y) to the axis of the GS X. A MEMS scanner with an oscillatory micro-mirror is similar with this schematics, but there is only one mirror that oscillates with regard to two perpendicular axis. In this respect, all the following discussion and equations that are deduced for the 2D GS can also be applied for the MEMS scanner, by considering no distance between the oscillatory axis, therefore d equals zero.

In Fig. 7(b) and (c) the two most utilized raster scanning algorithms are described [43,44], and they are detailed in Sections 5.3 and 5.4, respectively.

5.2. Scanning function of GS X – for the fast scan axis

On the x -axis, for the fast scan there are two possibilities that result from the discussion in the previous sections:

- A purely triangular scanning functions can be employed (as discussed in Section 2), with a triangular input. The output signal is a triangular-type function, with linear portion, but also with non-linear portions that are more extended in time, characterized by the 2τ time intervals – Fig. 1(b2), as the scan frequency f_s (therefore the scanning speed v) and the angular scan amplitude of the input signal, θ_m (or its corresponding linear one, H) – Fig. 1(a) – are increased.

- An optimized linear plus parabolic (p) scanning function [30], as developed in Section 3, that has pre-shaped linear and non-linear portions (the latter parabolic, for a maximum possible duty cycle). These non-linear portions are also characterized by certain 2τ time intervals (of the non-linear portions), although they are calculated differently than in the previous case, i.e. with Eq. (A.7).

In both cases the useful scan is performed linearly, with a constant scanning speed v of the laser spot on the “active” $2x_a$ portion, in a t_a time interval. For both cases the scanning functions – angular, $\theta(t)$, and linear, $x(t)$ – are presented in Fig. 8. Their graphs are consistent with Fig. 1(b2) – for the purely triangular input function, and also with Fig. 3 – for the linear plus parabolic (p) scanning function.

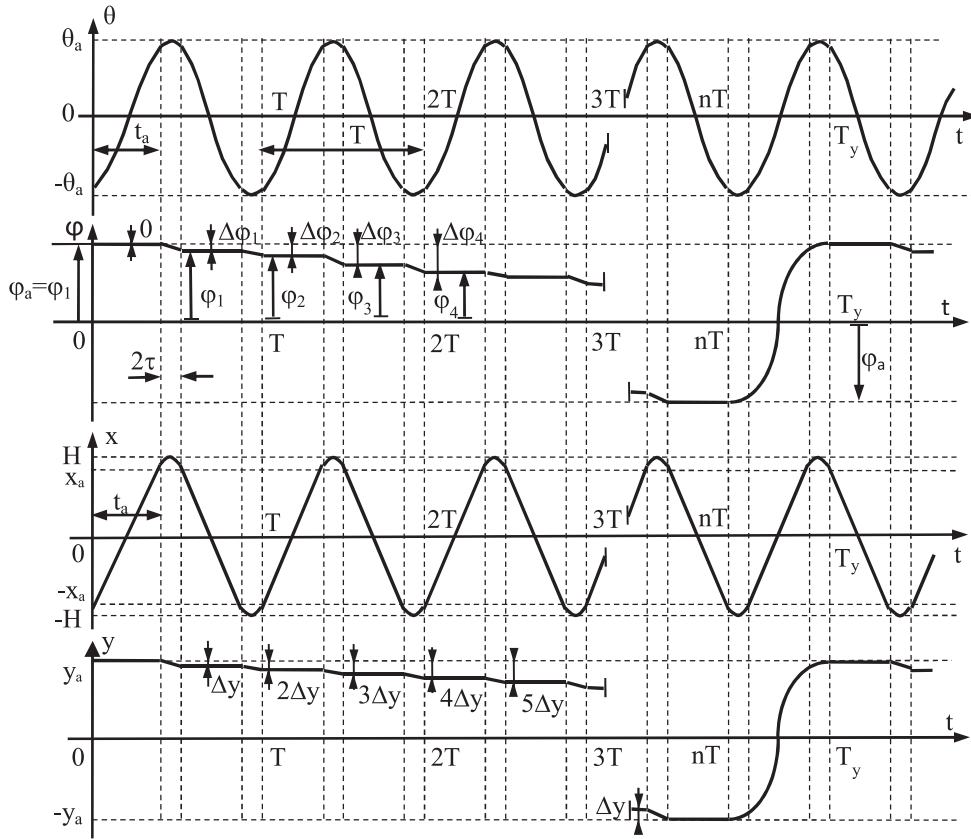


Fig. 8. 2D scanning functions – for both the fast scan axis Ox (angular function $\theta(t)$, and linear one $x(t)$) and the slow scan axis Oy (angular function $\varphi(t)$, and linear one $y(t)$ – the latter with a step-by-step movement).

The characteristic time intervals of the scanning process, as pointed out in Fig. 8, are: $T_x = T$, period of the $x(t)$ function; t_a , active time (that characterizes the linear portion) of the x -scan; nT , the time interval for the uni-directional scan of the entire y -axis (corresponding to the total displacement $n \cdot \Delta y$ of the spot in the vertical direction; T_y , time period of the function $y(t)$, which includes the returning $T_y - nT = t_a + 4\tau$ interval, from the final point B of the scan to the initial point A – Fig. 7(b) and (c).

As a remark, on the fast scan axis a PM scanning head can also be utilized, while a GS is also utilized for this slow axis, to position the scan line, as in Fig. 7(a). This is important as it extends the results of our study to PM plus GS 2D scanners, that gain momentum for laser manufacturing, for example, due to the high scan speeds provided by the PM on the fast scan axis. For such scanners, the PM theory recently developed provides the means to overcome the PM's main disadvantage, of the translation of the laser spot on the eccentric PM facet [15].

5.3. Step-by-step scanning function of the GS Y – for the slow scan axis

This first possible 2D scanning algorithm corresponds to a *step-by-step movement on the (slow scan) y -axis*, ideally with a constant step Δy on the y -axis, as presented in Fig. 7(b) and 8.

On the y -axis, in order to correlate the movements of the two 1D scanners, by imposing the constant steps Δy of the $y(t)$ function, the necessary angular scanning function $\varphi(t)$ has to be deduced. This scan resolution imposed on the y -axis determines the necessary number n of steps (therefore of $n + 1$ lines) that are necessary in order to scan the $2y_a$ distance – Fig. 7(b):

$$n = 2y_a / \Delta y. \quad (20)$$

In the system of coordinates in Figs. 7(a, b), and 8, the step-by-step positions on the y -axis are:

$$y_j = y_a - j \cdot \Delta y, \quad j = 0, n \quad (21)$$

where $y(t = 0) = y_a = -y(t = nT)$, as for simplicity the x -axis is considered to divide the scan area in two.

The corresponding angular positions φ_j (where $\varphi(t = 0)$ equals the angular amplitude φ_a of the GS Y) are deduced by taking into account that, for such constant steps on the y -axis, variable angular steps have to be used. Thus, from step “ j ”

to step “ $j + 1$ ”, as $\varphi_{j+1} = \varphi_j - \Delta\varphi$, one can write

$$\begin{cases} y_j = (L + d) \cdot \tan 2\varphi_j \\ y_{j+1} = y_j - \Delta y = (L + d) \cdot \tan 2(\varphi_j - \Delta\varphi_{j+1}) \end{cases}, j = \overline{0, n-1}, \quad (22)$$

therefore the variable angular step is

$$\Delta\varphi_{j+1} = \varphi_j - \frac{1}{2} \arctan \left(\tan 2\varphi_j - \frac{\Delta y}{L + d} \right), j = \overline{0, n-1}. \quad (23)$$

To pass from one step φ_j to the next one, φ_{j+1} (and, corresponding, from y_j to y_{j+1}), a portion that unites the two consecutive levels has to be imposed, for a time interval 2τ that corresponds to the non-linear portion of the x -axis scan, as discussed above.

Two methods exist in order to do this: (i) the simplest one seems to be with a portion that is scanned linearly with a constant speed on the y -axis; (ii) with a profile tangent to the two consecutive levels – and with a function that has to be written in order to comply with this condition.

(i) By choosing a constant linear speed $y(t)$ scanning function for the connecting portions, when passing from each step “ j ” to the step “ $j + 1$ ”, the spot has this linear speed:

$$v_{y \min} = \Delta y / 2\tau, t \in [t_j, t_{j+1}], j = \overline{0, n-1} \quad (24)$$

The equation of the lines that unite each step of the $y(t)$ graph is therefore

$$y_{j,j+1}(t) = v_{y \min}(-t + t_j) + y_j, t \in [t_j, t_{j+1}], j = \overline{0, n-1} \quad (25)$$

For the transition portions (from the “ j ” to the “ $j + 1$ ” step), the accelerations of the spot are $a_y(t_j) = -\infty$ and $a_y(t_j + 2\tau) = \infty$. This is however not an issue for the scanner, as these derivatives do not correspond to the actual oscillatory movement (which is the angular one), but to the apparent movement of the laser spot on a screen or on a lens.

In order to produce Eq. (25), from Fig. 7(a),

$$\varphi_{j,j+1}(t) = \frac{1}{2} \arctan \frac{v(t_j - t) + y_j}{L + d}, t \in [t_j, t_j + 2\tau], j = \overline{0, n-1}. \quad (26)$$

Its derivatives are

$$\begin{cases} \dot{\varphi}_{j,j+1}(t) = \frac{-(L + d)v_{y \min}/2}{(L + d)^2 + [v_{y \min}(t_j - t) + y_j]^2} \\ \ddot{\varphi}_{j,j+1}(t) = \frac{-(L + d)v_{y \min}^2[v_{y \min}(t_j - t) + y_j]}{[(L + d)^2 + [v_{y \min}(t_j - t) + y_j]^2]^2} \end{cases}, t \in [t_j, t_j + 2\tau], j = \overline{0, n-1}. \quad (27)$$

Significant particular values can be obtained from these equations for the angular velocity and acceleration of the GS Y:

$$\begin{cases} \omega_y(t_j) = \dot{\varphi}_{j,j+1}(t_j) = \frac{-(L + d)v_{y \min}}{2[(L + d)^2 + y_j^2]} \\ \omega_{yj} = \dot{\varphi}_{j,j+1}(t_j + \tau) = \frac{-(L + d)v_{y \min}}{2[(L + d)^2 + (y_j - \Delta y/2)^2]} \\ \omega_y(t_{j+1}) = \dot{\varphi}_{j,j+1}(t_j + 2\tau) = \frac{-(L + d)v_{y \min}}{2[(L + d)^2 + (y_j - \Delta y)^2]} \end{cases}, t \in [t_j, t_j + 2\tau], j = \overline{0, n-1} \quad (28)$$

and

$$\begin{cases} \varepsilon_y(t_j) = \ddot{\varphi}_{j,j+1}(t_j) = \frac{-(L + d)y_j v_{y \min}^2}{2[(L + d)^2 + y_j^2]^2} \\ \varepsilon_{yj} = \ddot{\varphi}_{j,j+1}(t_j + \tau) = \frac{-(L + d)(y_j - \Delta y/2)v_{y \min}^2}{2[(L + d)^2 + (y_j - \Delta y/2)^2]^2} \\ \varepsilon_y(t_{j+1}) = \ddot{\varphi}_{j,j+1}(t_j + 2\tau) = \frac{-(L + d)(y_j - \Delta y)v_{y \min}^2}{2[(L + d)^2 + (y_j - \Delta y)^2]^2} \end{cases}, t \in [t_j, t_j + 2\tau], j = \overline{0, n-1}. \quad (29)$$

Based on the trends in these particular values, the graphs in Fig. 9(a1) are represented. The above limits and maximum/peak values of the angular velocity ω_j and acceleration ε_j may seem fine and, because of the relative simplicity of

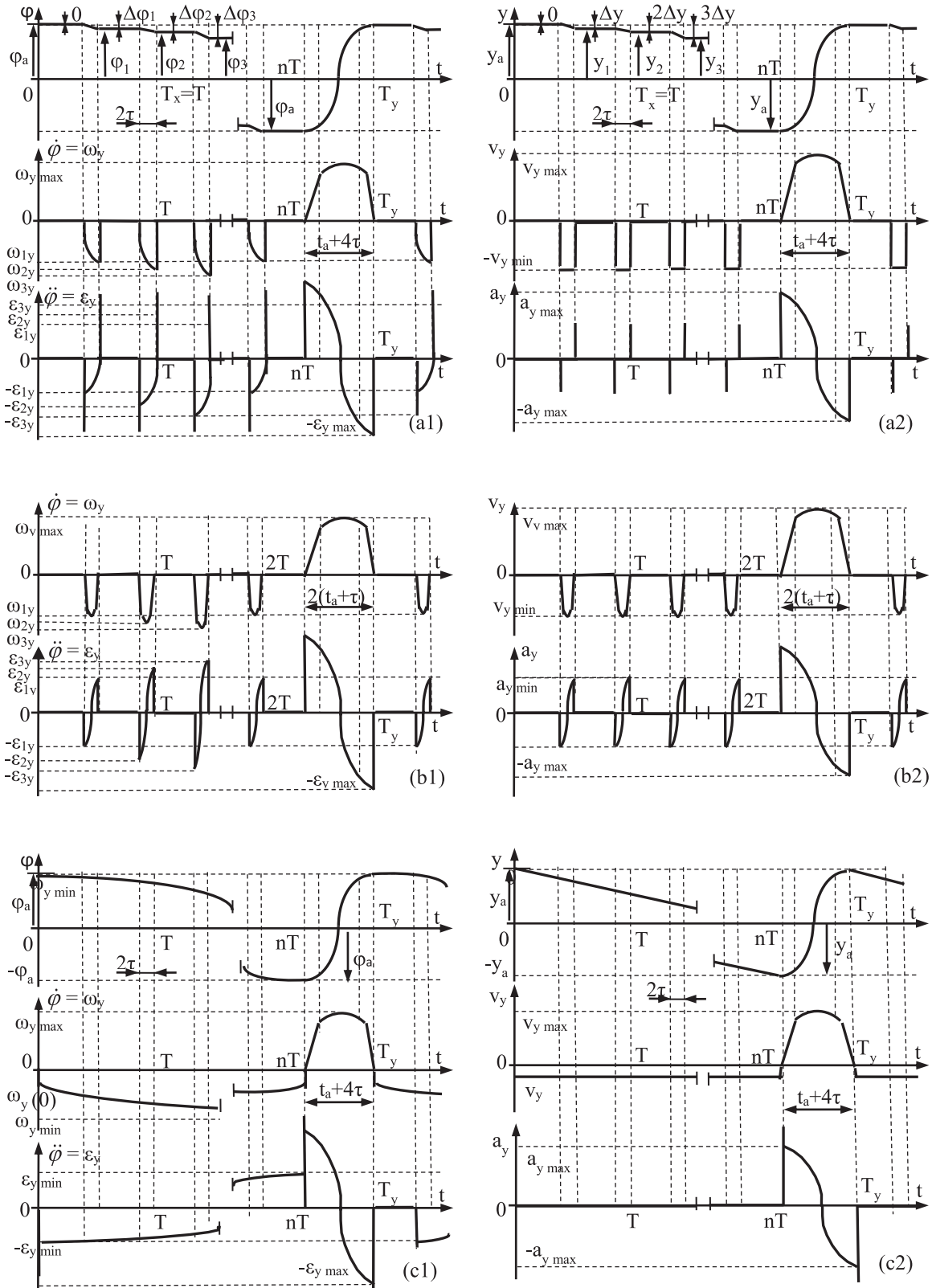


Fig. 9. Two types of 2D scanning functions, for the slow scan axis Oy , both (1) angular, $\varphi(t)$, and (2) linear, $y(t)$: for (a, b) a step-by-step movement, as well as for (c) a continuous movement with a constant scanning speed for the active nT portion of the scan. For the $y(t)$ step-by-step movement, two variants are considered for passing from one scan line to the following one: (a) with constant speed portions – case (i); (b) with a convenient equation – case (ii). Both the angular and the linear derivatives (velocity and acceleration) are deduced, for all the considered scanning functions.

this variant of the j line to the $j + 1$ line passing on the y -axis, this may look like the most appropriate one to be chosen. However, this is not true, as from Eq. (28) and Fig. 9(a1) a major issue can be noticed: because of the theoretically instantaneous increase and decrease of ω_j at each of the time moments t_j and t_{j+1} , respectively, a theoretically infinite angular acceleration ε_j occurs at these time moments. Because of the response time of the system, this does not actually happen [34], as it is well-known from automation, but one still have spikes of ε_j , therefore also of the inertia torque and of the input voltage that has to be applied, as discussed in Sections 3 and 4. In conclusion, these scanning functions should be avoided, if possible, and another scanning strategy is required, as developed in the following.

(ii) By choosing a convenient profile of the curve $\varphi(t)$ for the connecting portions: this curve may have the equation

$$\varphi_{j,j+1}(t) = \varphi_j - \frac{\Delta\varphi_j}{2} \cdot \left[1 - \cos \frac{\pi(t-t_j)}{2\tau} \right], t \in [t_j, t_j + 2\tau], j = \overline{0, n-1} \quad (30)$$

that provides a smooth transition from the “ j ” to the “ $j + 1$ ” horizontal scan line, with tangent curves to each of these scan lines, therefore with no angular acceleration spikes, as demonstrated bellow. This equation also satisfies the conditions $\varphi_{j,j+1}(t_j) = \varphi_j$ and $\varphi_{j,j+1}(t_j + 2\tau) = \varphi_{j+1}$ (Fig. 8).

The derivatives of this function are, from Eq. (24) and as represented in Figs. 8 and 9(b1),

$$\begin{cases} \dot{\varphi}_{j,j+1}(t) = \omega_y(t) = -\frac{\pi \cdot \Delta\varphi_{j+1}}{4\tau} \sin \frac{\pi(t-t_j)}{2\tau} \\ \ddot{\varphi}_{j,j+1}(t) = \varepsilon_y(t) = -\frac{\pi^2 \cdot \Delta\varphi_{j+1}}{8\tau^2} \cos \frac{\pi(t-t_j)}{2\tau} \end{cases}, t \in [t_j, t_j + 2\tau] \quad (31)$$

The extreme values of the angular velocity and acceleration on the GS Y can thus be obtained for this case:

$$\begin{cases} \dot{\varphi}_{j,j+1}(t_j) = \dot{\varphi}_{j,j+1}(t_{j+1}) = 0 \\ \omega_{jy}(t_j + \tau) = \dot{\varphi}_{j,j+1}(t_j + \tau) = -\frac{\pi \cdot \Delta\varphi_{j+1}}{4\tau} \\ \varepsilon_{jy}(t_j) = -\ddot{\varphi}_{j,j+1}(t_j) = \frac{\pi^2 \cdot \Delta\varphi_{j+1}}{8\tau^2} = \ddot{\varphi}_{j,j+1}(t_{j+1}) \\ \ddot{\varphi}_{j,j+1}(t_j + \tau) = 0 \end{cases}, t \in [t_j, t_{j+1}], j = \overline{0, n-1} \quad (32)$$

where $\Delta\varphi_j$ is given by Eq. (23). One can see that both $|\omega_{jy}|$ and $|\varepsilon_{jy}|$, therefore the peaks of both the angular velocity and acceleration increase as “ j ” progresses towards $n/2$, therefore towards the Ox axis, and then decrease again, as the scan lines are drawing further away from it, because it can be easily demonstrated from Eq. (23) that, in general, $\Delta\varphi_j < \Delta\varphi_{j+1}$, $j = \overline{0, n-1}$, therefore, in particular, $\Delta\varphi_1 < \Delta\varphi_{n/2} = \Delta\varphi_{n/2+1} < \Delta\varphi_n$.

Also, from the Eq. (32)₁, there is no sudden increase (or decrease) in the angular velocity at $t = t_j$ or at $t = t_{j+1}$, as the $\varphi_{j,j+1}(t)$ portions were chosen tangent to the two consecutive scan lines. The issue that appears in the case (i) has thus been avoided. On the other hand, the maximum values of the angular acceleration that are obtained in Eq. (32)₃ still have to be analyzed and compared with the other acceleration peaks in the total scan period T_y (see Section 5.6).

The transition portion from the “ j ” to the “ $j + 1$ ” horizontal scan line on the y -axis is, from Figs. 7 and 9(b2),

$$y_{j,j+1}(t) = (L + d) \tan 2\varphi_{j,j+1}(t), t \in [t_j, t_{j+1}], j = \overline{0, n-1} \quad (33)$$

and its derivatives are

$$\begin{cases} \dot{y}_{j,j+1}(t) = (L + d) \frac{2 \cdot \dot{\varphi}_{j,j+1}(t)}{\cos^2 \varphi_{j,j+1}(t)} \\ \ddot{y}_{j,j+1}(t) = 2(L + d) \frac{\ddot{\varphi}_{j,j+1}(t) \cdot \cos \varphi_{j,j+1}(t) + 2\dot{\varphi}_{j,j+1}^2(t) \cdot \sin \varphi_{j,j+1}(t)}{\cos^3 \varphi_{j,j+1}(t)} \end{cases}, t \in [t_j, t_{j+1}], j = \overline{0, n-1} \quad (34)$$

The above two equations can be easily verified: from Eq. (33), at the margins of the time interval, Eq. (22) are obtained; from Eq. (34),

$$\begin{cases} \dot{y}_{j,j+1}(t_j) = \dot{y}_{j,j+1}(t_{j+1}) = 0 \\ \dot{y}_{j,j+1}(t_j + \tau) = \frac{-\pi(L+d) \cdot \Delta\varphi_{j+1}}{2\tau \cos^2(\pi \cdot \Delta\varphi_{j+1}/4\tau)} \\ -\ddot{y}_{j,j+1}(t_j) = \ddot{y}_{j,j+1}(t_{j+1}) = \frac{-\pi^2(L+d) \cdot \Delta\varphi_{j+1}}{4\tau^2 \cos^2(\pi^2 \cdot \Delta\varphi_{j+1}/8\tau^2)} \\ \ddot{y}_{j,j+1}(t_j + \tau) = \frac{-\pi^2(L+d) \cdot (\Delta\varphi_{j+1})^2}{4\tau^2 \cos^2\varphi_{j,j+1}(t_j + \tau)} \tan \varphi_{j,j+1}(t_j + \tau) \end{cases}, t \in [t_j, t_j + 2\tau], j = \overline{0, n-1} \quad (35)$$

The peak of the linear scanning speed was thus obtained at $t = t_j + \tau$, but also the fact that the graph of the linear acceleration is not symmetrical, as the zero value is not obtained at the center of each time interval, although the values at the margins of each such interval are equal in modulus.

5.4. Continuous scanning function of the GS Y – for the slow scan axis

The second 2D scanning algorithm on the slow axis – Fig. 7(c) – corresponds to a continuous movement on the y-axis, with a constant speed, that from Fig. 9(c2) is

$$v_y = -2y_a/nT, \quad (36)$$

The equation of this movement on the y-axis graph is therefore

$$y(t) = v_y t + y_a, t \in [0, nT] \quad (37)$$

For the angular movement of the mirror of the GS Y that produces this movement of the spot, the equation that can be written from Fig. 7(c) is

$$\varphi(t) = \frac{1}{2} \arctan \frac{y(t)}{L+d}, t \in [0, nT], \quad (38)$$

and its derivatives are, similar to Eq. (27),

$$\begin{cases} \dot{\varphi}(t) = \frac{(L+d)v_y/2}{(L+d)^2 + (v_y t + y_a)^2} \\ \ddot{\varphi}(t) = \frac{-(L+d)v_y^2(v_y t + y_a)}{[(L+d)^2 + (v_y t + y_a)^2]^2} \end{cases}, t \in [0, nT]. \quad (39)$$

Significant particular values can be obtained from these equations for the angular velocity and acceleration of the GS Y:

$$\begin{cases} \omega_y(0) = \dot{\varphi}(0) = \frac{(L+d)v_y}{2[(L+d)^2 + y_a^2]} = \omega_y(nT) \\ \omega_{y \min} = \ddot{\varphi}(nT/2) = v_y/2(L+d) \end{cases} \quad (40)$$

and

$$\begin{cases} \varepsilon_{y \min} = \ddot{\varphi}(0) = \frac{-(L+d)y_a v_y^2}{[(L+d)^2 + y_a^2]^2} = \varepsilon_y(nT) \\ \varepsilon_y(nT/2) = 0 \end{cases}. \quad (41)$$

One can see that this case has the same issue like the one for case (i) in Section 5.3, but this issue cannot be avoided in this case, as the scanning algorithm implies a constant scanning speed on the y-axis. Therefore, a high (theoretically infinite) spike of the angular acceleration, of the inertia torque, and of the voltage that has to be applied to the systems appears for both $t = 0$ and $t = nT$.

One may therefore conclude that this scanning algorithm is less advantageous, and that the step-by-step scanning on the slow scan axis (in Fig. 7(a), with the GS Y) is preferable.

5.5. Returning portion of the scanning function of the GS Y

The returning portion of the laser spot from the final point B of the raster scanning to the initial point A is the same for all variants of the above scanning function – Figs. 7(b,c), 8, and 9. As for the other portions of this function, the equation of this returning part can be written in two ways, but the one with a constant speed movement for $y(t)$ produces spikes of the angular acceleration, therefore it is not a satisfactory solution.

One must therefore consider, similar to case (ii), [Section 5.3](#), a convenient equation for the angular movement of the galvomirror of the GS Y after the completion of a scan of the entire plane surface:

$$\varphi_{B-A}(t) = -\varphi_a \cdot \cos \frac{\pi(t - nT)}{t_a + 4\tau}, \quad t \in [nT, T_y]. \quad (42)$$

function written in order to have, as in [Figs. 8](#) and [9](#), $\varphi_{B-A}(nT) = -\varphi_a = \varphi_{B-A}(T_y)$ and $\varphi_{B-A}((nT + T_y)/2) = 0$. Also, as it can be deduced using the derivatives bellow, the graph of this function is tangent to the initial and to the final scan line – [Fig. 9](#).

The derivatives of this return function are, for the angular scanning – [Fig. 9\(a1, b1, c1\)](#):

$$\begin{cases} \dot{\varphi}_{B-A}(t) = \frac{\pi \cdot \varphi_a}{t_a + 4\tau} \sin \frac{\pi(t - nT)}{t_a + 4\tau} \\ \ddot{\varphi}_{B-A}(t) = \frac{\pi^2 \cdot \varphi_a}{(t_a + 4\tau)^2} \cos \frac{\pi(t - nT)}{t_a + 4\tau} \end{cases}, \quad t \in [nT, T_y] \quad (43)$$

The extreme values of the angular velocity and acceleration of the GS Y for this return time interval are therefore

$$\begin{cases} \omega_{y \max}(t) = \dot{\varphi}_{B-A}(nT + t_a/2 + 2\tau) = \frac{\pi \cdot \varphi_a}{t_a + 4\tau} \\ \varepsilon_{y \max}(t) = \ddot{\varphi}_{B-A}(nT) = \frac{\pi^2 \cdot \varphi_a}{(t_a + 4\tau)^2} = -\ddot{\varphi}_{B-A}(T_y) \end{cases} \quad (44)$$

The linear scanning functions on the y-axis for this return portion is, from [Fig. 7](#),

$$y_{B-A}(t) = (L + d) \tan 2\varphi_{B-A}(t), \quad (45)$$

therefore its derivatives, with the graphs shown in [Fig. 9\(a2, b2, c2\)](#), are

$$\begin{cases} \dot{y}_{B-A}(t) = v_y(t) = (L + d) \frac{2 \cdot \dot{\varphi}_{B-A}(t)}{\cos^2 \varphi_{B-A}(t)} \\ \ddot{y}_{B-A}(t) = a_y(t) = 2(L + d) \frac{\ddot{\varphi}_{B-A}(t) \cdot \cos \varphi_{B-A}(t) + 2\dot{\varphi}_{B-A}(t) \cdot \sin \varphi_{B-A}(t)}{\cos^3 \varphi_{B-A}(t)} \end{cases} \quad (46)$$

In order to verify the above two equations, by using them we obtain: $y_{B-A}(nT) = -y_a = y_{B-A}(T_y)$, $v_y(nT) = \dot{y}_{B-A}(nT) = \dot{y}_{B-A}(T_y) = 0$, and $a_y(nT + t_a/2 + 2\tau) = 0$.

The maximum values of the velocity and acceleration for this time portion are, from [Eq. \(46\)](#),

$$\begin{cases} v_{y \max} = \dot{y}_{B-A}(nT + t_a/2 + 2\tau) = \frac{2\pi(L + d) \cdot \varphi_a}{t_a + 4\tau} \\ a_{y \max} = \ddot{y}_{B-A}(nT + t_a/2 + 2\tau) = \frac{2\pi^2(L + d) \cdot \varphi_a}{(t_a + 4\tau)^2 \cos^2 \varphi_a} \end{cases} \quad (47)$$

5.6. Maximum angular accelerations for the optimal scanning function on the slow scan axis

A final aspect that has to be determined in order to conclude the discussion is related to the maximum values of the angular acceleration of the GS Y (on the slow scan axis, [Fig. 7](#)). This is important, as these angular acceleration values determine, as discussed in [Section 4](#), the peaks of the inertia torque, thus of the voltage inputs that have to be applied to the device in order to produce a certain scanning function of the GS.

For the optimal, step-by-step scanning function given by [Eq. \(30\)](#), the angular acceleration was obtained in [Eq. \(31\)₂](#). From the acceleration maximum values of each time interval in [Eq. \(32\)₃](#), by taking into account the angular steps given by [Eq. \(23\)](#) – and the corresponding discussion, the maximum value of the angular acceleration on the step-by-step portion are obtained before and after the (scan line that is placed on the) optical axis (O.A.) of the scanner ([Fig. 7](#)), as pointed out in [Fig. 9](#):

$$\varepsilon_{j \max} = \varepsilon_{n/2y} = \ddot{\varphi}_{n/2-1, n/2}(nT/2) = \ddot{\varphi}_{n/2, n/2+1}((n+1)T/2) = \frac{\pi^2 \cdot \Delta\varphi_{n/2}}{8\tau^2}, \quad (48a)$$

therefore

$$\varepsilon_{j \max} = \frac{\pi^2}{16\tau^2} \arctan \frac{\Delta y}{L + d}, \quad (48b)$$

For the returning portion of the scanning function, [Eq. \(42\)](#), the angular acceleration was obtained in [Eq. \(43\)₂](#), therefore the acceleration peak for this portion is given by [Eq. \(44\)₂](#), [Fig. 9](#):

$$\varepsilon_{\max}^{B-A} = \ddot{\varphi}_{B-A}(nT) = \frac{\pi^2 \cdot \varphi_a}{(T + 2\tau)^2}. \quad (49)$$

A final comparison can be made between the two values of the acceleration in Eqs. (48) and (49). Thus,

$$\varepsilon_{\max}^{B-A} > \varepsilon_{j\max} \quad \Leftrightarrow \quad \Delta y < (L + d) \tan \left[\left(\frac{4\tau}{T/2 + 2\tau} \right)^2 \varphi_a \right] = \Delta y_{\min}. \quad (50)$$

For imaging, but also for most applications, Δy has to be as small as possible (in order to increase resolution), therefore the conclusion of this comparison is: for a linear resolution better than the one corresponding to the threshold $\Delta y_{\min}$ defined by Eq. (50), the maximum angular amplitude reached in the return portion B–A of the scanning function of the GS Y, Eq. (49) is the most significant, and it is the one that has to be taken into account in the discussion of the voltage peaks of the GS Y.

6. Conclusions

1D and 2D GSs and MEMS with oscillatory mirrors were studied from the point of view of their scanning functions. The behavior of these laser systems, with their specific optical requirements (especially regarding their necessary scanning speed, but also FOV), was studied from the point of view of their mechanical and electrical issues. The most common input and output signals (sinusoidal, sawtooth, and triangular) were considered – from our experimental studies on scanning in biomedical imaging, in particular in OCT [26,28]. The duty cycle/time efficiency and distortions of these scanning regimes for 1D scanners were discussed. Further on we determined the optimal scanning functions of a GS or MEMS scanner, linear (on the active portions, swept with a constant scanning speed) and parabolic (on the stop-and-turn, non-linear portions). We demonstrated theoretically that such functions (and not the linear plus sinusoidal functions pointed out in the literature [7]) produce the maximum duty cycle of the device [30], not only in open loop, but also in close loop (from the automation point of view of the scanners). Several methods that can be applied in order to decrease the peaks of the input voltage applied to the scanner are also proposed based on the equations of this optimal linear plus parabolic function. This is necessary in order to protect the systems from an electrical point of view – with a minimum loss in terms of the duty cycle.

2D scanning functions (for both 2D GSs or MEMS with oscillatory mirrors) were developed, utilizing one of the 1D scanning functions discussed above for the fast scan axis. The two most common scanning algorithms from the point of view of the slow scan axis were considered: (1) with a step-by-step movement and (2) with a continuous, constant scanning speed movement – Fig. 7(b) and (c), respectively. The scanning functions were obtained and analyzed for this slow scan axis, with their possible variants and equations. The optimal variant and scanning function was chosen from this analysis for the scanning algorithm (1). We also demonstrated the this algorithm (1) is better than (2), as it allows for avoiding to produce spikes of the angular acceleration, inertia torque, and voltage for the scanner. Therefore a rigorous demonstration of the fact that step-by-step scanning should be employed on the slow scan axis was made. *These conclusions, i.e. optimal 1D and 2D scanning functions can be applied for any laser scanners with oscillatory elements, including piezoelectric or of another type.*

The conclusions on the optimal scanning functions for the slow scan axis are also applicable to PM plus GS 2D scanners [15,45,46], which use the GS for this slow scan axis, as in our study.

Future work in this direction includes experimental work with the different 2D scanning functions developed in this study in a comparative look, for OCT (both for biomedical imaging and for NDT), but also for other applications, such as laser manufacturing. Handheld scanning probes for OCT, equipped with both 2D GSs and MEMS were developed in our group [37], but also by numerous others [35,36,38–40], and their testing and optimization are another direction of work in this respect. A comparison of the raster scanning approached in this study is also necessary with other systems, especially with Risley prisms [16–18], but also with other scanning modalities, such as alpha-beta scanning [47].

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Appendix A. Linear plus parabolic scanning function (p)

(i) For $t \in [0, t_a]$, $x(t)$ has a first linear portion, of equation and conditions – Fig. 3(b):

$$x = at + b, \text{ with } x(0) = -x_a; \quad x(t_a/2) = 0 \text{ and } \dot{x}(t) = v = \text{cst.}, \quad (A.1)$$

therefore

$$x(t) = vt - x_a, \quad (A.2)$$

where $v = 2x_a/t_a$ is the scanning velocity, which is a designing equation for this case.

(ii) For $t \in [t_a, T/2]$, a parabolic function is considered for this portion of the graph:

$$x(t) = at^2 + bt + c \Rightarrow \dot{x} = 2at + b, \quad (\text{A.3})$$

where a , b , and c are constants to be determined from the continuity and smoothness conditions of the $x(t)$ scanning function (Fig. 3):

$$\dot{x}(t_a + \tau) = 0; x(t_a) = x(T/2) = x_a; \dot{x}(t_a) = -\dot{x}(T/2) = v. \quad (\text{A.4})$$

From Eqs. (A.3) and (A.4), the coefficients are:

$$a = -v/2\tau; b = v(t_a + \tau)/\tau; c = -x_a(t_a + \tau)/\tau. \quad (\text{A.5})$$

and a designing equation is also obtained:

$$Ht_a = x_a(t_a + \tau) \Leftrightarrow v = 2H/(t_a + \tau) \quad (\text{A.6})$$

(iii) For $t \in [T/2, T-2\tau]$, the function $x(t)$ is linear, Eq. (A.1), but with the conditions – Fig. 3(b):

$$x(T/2) = x_a; x(T - 2\tau) = -x_a; \dot{x}(t) = v, \quad (\text{A.7})$$

therefore

$$x(t) = -vt + \frac{3t_a + 4\tau}{t_a}x_a. \quad (\text{A.8})$$

(iv) For $t \in [T-2\tau, T]$ the parabolic function, Eq. (A.1) has the conditions – Fig. 3(b):

$$\dot{x}(T - \tau) = 0, x(T - 2\tau) = x(T) = x_a; \dot{x}(T) = -\dot{x}(T - 2\tau) = v, \quad (\text{A.9})$$

therefore its coefficients are

$$a = v/2\tau; b = -v(T - \tau)/\tau; c = x_a + vt(T - 2\tau)/2\tau. \quad (\text{A.10})$$

The designing equation (A.6) is obtained in this case, as well.

Appendix B. Linear plus sinusoidal scanning function (s)

(i) and (iii) For $t \in [0, t_a]$ and $t \in [T/2, T-2\tau]$, the linear functions (A.2) and (A.8), respectively are valid for this case, as well.

(ii) For $t \in [t_a, T/2]$, a sinusoidal function is considered for the non-linear portion of $x(t)$ in this case:

$$x(t) = a \sin(\omega t + \varphi) \Rightarrow \dot{x}(t) = a\omega \cos(\omega t + \varphi). \quad (\text{B.1})$$

The simplest way to solve this case is to consider $x(t)$ with $x(t = 0) = H$ and then to make a change of variables. For a function translated with the origin in $x(t = 0) = H$ and given by Eq. (B.1), with the conditions obtained from Fig. 3(b):

$$x(0) = H; x(\tau) = x_a; \dot{x}(0) = 0; \dot{x}(\tau) = v, \quad (\text{B.2})$$

its coefficients are

$$a = H; \varphi = \pi/2. \quad (\text{B.3})$$

and its angular frequency is

$$\omega = \frac{v}{\sqrt{H^2 - x_a^2}}. \quad (\text{B.4})$$

From the other relation obtained, $\tan \omega\tau = v/\omega x_a$, the stop-and-turn portion is characterized by the half-time interval:

$$\tau = \frac{\sqrt{H^2 - x_a^2}}{v} \arctan \frac{\sqrt{H^2 - x_a^2}}{x_a}. \quad (\text{B.5})$$

In conclusion, from Eq. (B.1), using Eqs. (B.3) and (B.4), the translated sinusoidal function is:

$$x(t) = H \sin(\omega t + \pi/2). \quad (\text{B.6})$$

Therefore, the actual sinusoidal portion of the scanning function (i.e., the one that starts at the beginning of the active portion, as shown in Fig. 3(b) for the (p) function) is obtained using the change of variables:

$$t' = t - (t_a + 3\tau) \quad (\text{B.7})$$

Finally, by replacing Eq. (B.7) in (B.6), and by returning to the notation in t , the actual sinusoidal portion is:

$$x(t) = H \sin[\omega(t + t_a + 3\tau) + \pi/2]. \quad (\text{B.8})$$

(iv) For $t \in [T - 2\tau, T]$, the same procedure as for portion (ii) is applied: $x(t)$ also has Eq. (B.1), and with conditions (A.9), its coefficients are

$$a = -H; \varphi = \pi/2 - \omega T/2, \quad (\text{B.9})$$

while ω and τ are also given by Eqs. (B.3) and (B.5), respectively. The necessary change of variables is in this case:

$$t' = t - (t_a + 3\tau - T/2) \quad (\text{B.10})$$

Finally, by replacing Eq. (B.7) in (B.6), and by returning to the notation in t , the actual sinusoidal portion is:

$$x(t) = -H \sin[\omega(t - \tau) + \pi/2]. \quad (\text{B.11})$$

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